

Bayesian Reasoning and Machine Learning

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Notation List

$\mathcal{V}$	a calligraphic symbol typically denotes a set of random variables 3
$\text{dom}(x)$	Domain of a variable 3
$x = \mathbf{x}$	The variable x is in the state $\mathbf{x}$ 3
$p(x = \mathbf{tr})$	probability of event/variable x being in the state true 3
$p(x = \mathbf{fa})$	probability of event/variable x being in the state false 3
$p(x, y)$	probability of x and y 4
$p(x \cap y)$	probability of x and y 4
$p(x \cup y)$	probability of x or y 4
$p(x y)$	The probability of x conditioned on y 4
$\int_x f(x)$	For continuous variables this is shorthand for $\int f(x)dx$ and for discrete variables means summation over the states of x , $\sum_x f(x)$ 7
$\mathbb{I}[x = y]$	Indicator : has value 1 if $x = y$, 0 otherwise 11
$\text{pa}(x)$	The parents of node x 19
$\text{ch}(x)$	The children of node x 19
$\text{ne}(x)$	Neighbours of node x 20
$\mathcal{X} \perp\!\!\!\perp \mathcal{Y} \mathcal{Z}$	Variables $\mathcal{X}$ are independent of variables $\mathcal{Y}$ conditioned on variables $\mathcal{Z}$. 33
$\mathcal{X} \Pi \mathcal{Y} \mathcal{Z}$	Variables $\mathcal{X}$ are dependent on variables $\mathcal{Y}$ conditioned variables $\mathcal{Z}$ 33
$\dim x$	For a discrete variable x , this denotes the number of states x can take .. 43
$\langle f(x) \rangle_{p(x)}$	The average of the function $f(x)$ with respect to the distribution $p(x)$. 139
$\delta(a, b)$	Delta function. For discrete a, b , this is the Kronecker delta, $\delta_{a,b}$ and for continuous a, b the Dirac delta function $\delta(a - b)$ 142
$\dim \mathbf{x}$	The dimension of the vector/matrix $\mathbf{x}$ 150
$\sharp(x = s, y = t)$	The number of times variable x is in state s and y in state t simultaneously. 172
$\mathcal{D}$	Dataset 251
n	Data index 251
N	Number of Dataset training points 251
$\sharp_y^x$	The number of times variable x is in state y 265
S	Sample Covariance matrix 283
$\sigma(x)$	The logistic sigmoid $1/(1 + \exp(-x))$ 319
$\text{erf}(x)$	The (Gaussian) error function 319
$i \sim j$	The set of unique neighbouring edges on a graph 529
$\mathbf{I}_m$	The $m \times m$ identity matrix 546

Machine Learning

The last decade has seen considerable growth in interest in Artificial Intelligence and Machine Learning. In the broadest sense, these fields aim to ‘learn something useful’ about the environment within which the organism operates. How gathered information is processed leads to the development of algorithms – how to process high dimensional data and deal with uncertainty. In the early stages of research in Machine Learning and related areas, similar techniques were discovered in relatively isolated research communities. Whilst not all techniques have a natural description in terms of probability theory, many do, and it is the framework of Graphical Models (a marriage between graph and probability theory) that has enabled the understanding and transference of ideas from statistical physics, statistics, machine learning and information theory. To this extent it is now reasonable to expect that machine learning researchers are familiar with the basics of statistical modelling techniques.

This book concentrates on the probabilistic aspects of information processing and machine learning. Certainly no claim is made as to the correctness or that this is the only useful approach. Indeed, one might counter that this is unnecessary since “biological organisms don’t use probability theory”. Whether this is the case or not, it is undeniable that the framework of graphical models and probability has helped with the explosion of new algorithms and models in the machine learning community. One should also be clear that Bayesian viewpoint is not the only way to go about describing machine learning and information processing. Bayesian and probabilistic techniques really come into their own in domains where uncertainty is a necessary consideration.

The structure of the book

One aim of part I of the book is to encourage Computer Science students into this area. A particular difficulty that many modern students face is a limited formal training in calculus and linear algebra, meaning that minutiae of continuous and high-dimensional distributions can turn them away. In beginning with probability as a form of reasoning system, we hope to show the reader how ideas from logical inference and dynamical programming that they may be more familiar with have natural parallels in a probabilistic context. In particular, Computer Science students are familiar with the concept of algorithms as core. However, it is more common in machine learning to view the model as core, and how this is implemented is secondary. From this perspective, understanding how to translate a mathematical model into a piece of computer code is central.

Part II introduces the statistical background needed to understand continuous distributions and how learning can be viewed from a probabilistic framework. Part III discusses machine learning topics. Certainly some readers will raise an eyebrow to see their favourite statistical topic listed under machine learning. A difference viewpoint between statistics and machine learning is what kinds of systems we would ultimately

like to construct (machines capable of ‘human/biological information processing tasks’) rather than in some of the techniques. This section of the book is therefore what I feel would be useful for machine learners to know.

Part IV discusses dynamical models in which time is explicitly considered. In particular the Kalman Filter is treated as a form of graphical model, which helps emphasise what the model is, rather than focusing on it as a ‘filter’, as is more traditional in the engineering literature.

Part V contains a brief introduction to approximate inference techniques, including both stochastic (Monte Carlo) and deterministic (variational) techniques.

The references in the book are not generally intended as crediting authors with ideas, nor are they always to the most authoritative works. Rather, the references are largely to works which are at a level reasonably consistent with the book and which are readily available.

Whom this book is for

My primary aim was to write a book for final year undergraduates and graduates without significant experience in calculus and mathematics that gave an inroad into machine learning, much of which is currently phrased in terms of probabilities and multi-variate distributions. The aim was to encourage students that apparently unexciting statistical concepts are actually highly relevant for research in making intelligent systems that interact with humans in a natural manner. Such a research programme inevitably requires dealing with high-dimensional data, time-series, networks, logical reasoning, modelling and uncertainty.

Other books in this area

Whilst there are several excellent textbooks in this area, none currently meets the requirements that I personally need for teaching, namely one that contains demonstration code and gently introduces probability and statistics before leading on to more advanced topics in machine learning. This lead me to build on my lecture material from courses given at Aston, Edinburgh, EPFL and UCL and expand the demonstration software considerably. The book is due for publication by Cambridge University Press in 2010.

The literature on machine learning is vast, as is the overlap with the relevant areas of statistics, engineering and other physical sciences. In this respect, it is difficult to isolate particular areas, and this book is an attempt to integrate parts of the machine learning and statistics literature. The book is written in an informal style at the expense of rigour and detailed proofs. As an introductory textbook, topics are naturally covered to a somewhat shallow level and the reader is referred to more specialised books for deeper treatments. Amongst my favourites are:

- Graphical models
 - *Graphical models* by S. Lauritzen, Oxford University Press, 1996.
 - *Bayesian Networks and Decision Graphs* by F. Jensen and T. D. Nielsen, Springer Verlag, 2007.
 - *Probabilistic Networks and Expert Systems* by R. G. Cowell, A. P. Dawid, S. L. Lauritzen and D. J. Spiegelhalter, Springer Verlag, 1999.
 - *Probabilistic Reasoning in Intelligent Systems* by J. Pearl, Morgan Kaufmann, 1988.
 - *Graphical Models in Applied Multivariate Statistics* by J. Whittaker, Wiley, 1990.
 - *Probabilistic Graphical Models: Principles and Techniques* by D. Koller and N. Friedman, MIT Press, 2009.
- Machine Learning and Information Processing
 - *Information Theory, Inference and Learning Algorithms* by D. J. C. MacKay, Cambridge University Press, 2003.

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- *Pattern Recognition and Machine Learning* by C. M. Bishop, Springer Verlag, 2006.
 - *An Introduction To Support Vector Machines*, N. Cristianini and J. Shawe-Taylor, Cambridge University Press, 2000.
 - *Gaussian Processes for Machine Learning* by C. E. Rasmussen and C. K. I. Williams, MIT press, 2006.

How to use this book

Part I would be suitable for an introductory course on Graphical Models with a focus on inference. Part II contains enough material for a short lecture course on learning in probabilistic models. Part III is reasonably self-contained and would be suitable for a course on Machine Learning from a probabilistic perspective, particularly combined with the dynamical models material in part IV. Part V would be suitable for a short course on approximate inference.

Accompanying code

The MATLAB code is provided to help readers see how mathematical models translate into actual code. The code is not meant to be an industrial strength research tool, rather a reasonably lightweight toolbox that enables the reader to play with concepts in graph theory, probability theory and machine learning. In an attempt to retain readability, no extensive error and/or exception handling has been included. The code contains at the moment basic routines for manipulating discrete variable distributions, along with a set of routines that are more concerned with continuous variable machine learning. One could in principle extend the ‘graphical models’ part of the code considerably to support continuous variables. Limited support for continuous variables is currently provided so that, for example, inference in the linear dynamical system may be written in conducted of operations on Gaussian potentials. However, in general, potentials on continuous variables need to be manipulated with care and often specialised routines are required to ensure numerical stability.

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A final thankyou to my family and friends.

Website

The code along with an electronic version of the book is available from

<http://www.cs.ucl.ac.uk/staff/D.Barber/brml>

Instructors seeking solutions to the exercises can find information at the website, along with additional teaching material. The website also contains a feedback form and errata list.

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Part I

Inference in Probabilistic Models

1.1 Probability Refresher

Variables, States and Notational Shortcuts

Variables will be denoted using either upper case X or lower case x and a set of variables will typically be denoted by a calligraphic symbol, for example $\mathcal{V} = \{a, B, c\}$.

The *domain* of a variable x is written $\text{dom}(x)$, and denotes the states x can take. States will typically be represented using sans-serif font. For example, for a coin c , we might have $\text{dom}(c) = \{\text{heads}, \text{tails}\}$ and $p(c = \text{heads})$ represents the probability that variable c is in state **heads**.

The meaning of $p(\text{state})$ will often be clear, without specific reference to a variable. For example, if we are discussing an experiment about a coin c , the meaning of $p(\text{heads})$ is clear from the context, being shorthand for $p(c = \text{heads})$. When summing (or performing some other operation) over a variable $\sum_x f(x)$, the interpretation is that all states of x are included, *i.e.* $\sum_x f(x) \equiv \sum_{s \in \text{dom}(x)} f(x = s)$.

For our purposes, *events* are expressions about random variables, such as *Two heads in 6 coin tosses*. Two events are *mutually exclusive* if they cannot both simultaneously occur. For example the events *The coin is heads* and *The coin is tails* are mutually exclusive. One can think of defining a new variable named by the event so, for example, $p(\text{The coin is tails})$ can be interpreted as $p(\text{The coin is tails} = \text{true})$. We use $p(x = \text{tr})$ for the probability of event/variable x being in the state **true** and $p(x = \text{fa})$ for the probability of event/variable x being in the state **false**.

The Rules of Probability

Definition 1 (Rules of Probability (Discrete Variables)).

The probability of an event x occurring is represented by a value between 0 and 1.

$p(x) = 1$ means that we are certain that the event does occur.

Conversely, $p(x) = 0$ means that we are certain that the event does not occur.

The summation of the probability over all the states is 1:

$$\sum_x p(x = x) = 1 \tag{1.1.1}$$

Such probabilities are normalised. We will usually more conveniently write $\sum_x p(x) = 1$.

Two events x and y can interact through

$$p(x \text{ or } y) = p(x) + p(y) - p(x \text{ and } y) \quad (1.1.2)$$

We will use the shorthand $p(x, y)$ for $p(x \text{ and } y)$. Note that $p(y, x) = p(x, y)$ and $p(x \text{ or } y) = p(y \text{ or } x)$.

Definition 2 (Set notation). An alternative notation in terms of set theory is to write

$$p(x \text{ or } y) \equiv p(x \cup y), \quad p(x, y) \equiv p(x \cap y) \quad (1.1.3)$$

Definition 3 (Marginals). Given a *joint distribution* $p(x, y)$ the distribution of a single variable is given by

$$p(x) = \sum_y p(x, y) \quad (1.1.4)$$

Here $p(x)$ is termed a *marginal* of the joint probability distribution $p(x, y)$. The process of computing a marginal from a joint distribution is called *marginalisation*. More generally, one has

$$p(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = \sum_{x_i} p(x_1, \dots, x_n) \quad (1.1.5)$$

An important definition that will play a central role in this book is conditional probability.

Definition 4 (Conditional Probability / Bayes' Rule). The probability of event x conditioned on knowing event y (or more shortly, the probability of x given y) is defined as

$$p(x|y) \equiv \frac{p(x, y)}{p(y)} \quad (1.1.6)$$

If $p(y) = 0$ then $p(x|y)$ is not defined.

Probability Density Functions

Definition 5 (Probability Density Functions). For a single continuous variable x , the probability density $p(x)$ is defined such that

$$p(x) \geq 0 \quad (1.1.7)$$

$$\int_{-\infty}^{\infty} p(x) dx = 1 \quad (1.1.8)$$

$$p(a < x < b) = \int_a^b p(x)dx \quad (1.1.9)$$

As shorthand we will sometimes write $\int_{x=a}^b p(x)$, particularly when we want an expression to be valid for either continuous or discrete variables. The multivariate case is analogous with integration over all real space, and the probability that x belongs to a region of the space defined accordingly.

For continuous variables, formally speaking, events are defined for the variable occurring within a defined region, for example

$$p(x \in [-1, 1.7]) = \int_{-1}^{1.7} f(x)dx \quad (1.1.10)$$

where here $f(x)$ is the probability density function (pdf) of the continuous random variable x . Unlike probabilities, probability densities can take positive values greater than 1.

Formally speaking, for a continuous variable, one should not speak of the probability that $x = 0.2$ since the probability of a single value is always zero. However, we shall often write $p(x)$ for continuous variables, thus not distinguishing between probabilities and probability density function values. Whilst this may appear strange, the nervous reader may simply replace our $p(X = x)$ notation for $\int_{x \in \Delta} f(x)dx$, where Δ is a small region centred on x . This is well defined in a probabilistic sense and, in the limit Δ being very small, this would give approximately $\Delta f(x)$. If we consistently use the same Δ for all occurrences of pdfs, then we will simply have a common prefactor Δ in all expressions. Our strategy is to simply ignore these values (since in the end only relative probabilities will be relevant) and write $p(x)$. In this way, all the standard rules of probability carry over, including Bayes' Rule.

Interpreting Conditional Probability

Imagine a circular dart board, split into 20 equal sections, labelled from 1 to 20 and Randy, a dart thrower who hits any one of the 20 sections uniformly at random. Hence the probability that a dart thrown by Randy occurs in any one of the 20 regions is $p(\text{region } i) = 1/20$. A friend of Randy tells him that he hasn't hit the 20 region. What is the probability that Randy has hit the 5 region? Conditioned on this information, only regions 1 to 19 remain possible and, since there is no preference for Randy to hit any of these regions, the probability is $1/19$. The conditioning means that certain states are now inaccessible, and the original probability is subsequently distributed over the remaining accessible states. From the rules of probability :

$$p(\text{region } 5 | \text{not region } 20) = \frac{p(\text{region } 5, \text{not region } 20)}{p(\text{not region } 20)} = \frac{p(\text{region } 5)}{p(\text{not region } 20)} = \frac{1/20}{19/20} = \frac{1}{19}$$

giving the intuitive result. In the above $p(\text{region } 5, \text{not region } 20) = p(\text{region } \{5 \cap 1 \cap 2 \cap \dots \cap 19\}) = p(\text{region } 5)$.

An important point to clarify is that $p(A = a | B = b)$ should not be interpreted as 'Given the event $B = b$ has occurred, $p(A = a | B = b)$ is the probability of the event $A = a$ occurring'. In most contexts, no such explicit temporal causality is implied¹ and the correct interpretation should be ' $p(A = a | B = b)$ is the probability of A being in state a under the constraint that B is in state b '.

The relation between the conditional $p(A = a | B = b)$ and the joint $p(A = a, B = b)$ is just a normalisation constant since $p(A = a, B = b)$ is not a distribution in A – in other words, $\sum_a p(A = a, B = b) \neq 1$. To make it a distribution we need to divide : $p(A = a, B = b) / \sum_a p(A = a, B = b)$ which, when summed over a does sum to 1. Indeed, this is just the definition of $p(A = a | B = b)$.

¹We will discuss issues related to causality further in section(3.4).

Definition 6 (Independence).

Events x and y are independent if knowing one event gives no extra information about the other event. Mathematically, this is expressed by

$$p(x, y) = p(x)p(y) \quad (1.1.11)$$

Provided that $p(x) \neq 0$ and $p(y) \neq 0$ independence of x and y is equivalent to

$$p(x|y) = p(x) \Leftrightarrow p(y|x) = p(y) \quad (1.1.12)$$

If $p(x|y) = p(x)$ for all states of x and y , then the variables x and y are said to be independent. If

$$p(x, y) = kf(x)g(y) \quad (1.1.13)$$

for some constant k , and positive functions $f(\cdot)$ and $g(\cdot)$ then x and y are independent.

Deterministic Dependencies

Sometimes the concept of independence is perhaps a little strange. Consider the following : variables x and y are both binary (their domains consist of two states). We define the distribution such that x and y are always both in a certain joint state:

$$\begin{aligned} p(x = \mathbf{a}, y = 1) &= 1 \\ p(x = \mathbf{a}, y = 2) &= 0 \\ p(x = \mathbf{b}, y = 2) &= 0 \\ p(x = \mathbf{b}, y = 1) &= 0 \end{aligned}$$

Are x and y dependent? The reader may show that $p(x = \mathbf{a}) = 1$, $p(x = \mathbf{b}) = 0$ and $p(y = 1) = 1$, $p(y = 2) = 0$. Hence $p(x)p(y) = p(x, y)$ for all states of x and y , and x and y are therefore independent. This may seem strange – we know for sure the relation between x and y , namely that they are always in the same joint state, yet they are independent. Since the distribution is trivially concentrated in a single joint state, knowing the state of x tells you nothing that you didn't anyway know about the state of y , and vice versa.

This potential confusion comes from using the term ‘independent’ which, in English, suggests that there is no influence or relation between objects discussed. The best way to think about statistical independence is to ask whether or not knowing the state of variable y tells you something more than you knew before about variable x , where ‘knew before’ means working with the joint distribution of $p(x, y)$ to figure out what we can know about x , namely $p(x)$.

1.1.1 Probability Tables

Based on the populations 60776238, 5116900 and 2980700 of England (E), Scotland (S) and Wales (W), the a priori probability that a randomly selected person from these three countries would live in England, Scotland or Wales, would be approximately 0.88, 0.08 and 0.04 respectively. We can write this as a vector (or probability table) :

$$\begin{pmatrix} p(Cnt = \mathbf{E}) \\ p(Cnt = \mathbf{S}) \\ p(Cnt = \mathbf{W}) \end{pmatrix} = \begin{pmatrix} 0.88 \\ 0.08 \\ 0.04 \end{pmatrix} \quad (1.1.14)$$

whose component values sum to 1. The ordering of the components in this vector is arbitrary, as long as it is consistently applied.

For the sake of simplicity, let's assume that only three Mother Tongue languages exist : English (**Eng**), Scottish (**Scot**) and Welsh (**Wel**), with conditional probabilities given the country of residence, England (**E**), Scotland (**S**) and Wales (**W**). We write a (fictitious) conditional probability table

$$\begin{aligned} p(MT = \text{Eng}|Cnt = E) &= 0.95 & p(MT = \text{Scot}|Cnt = E) &= 0.04 & p(MT = \text{Wel}|Cnt = E) &= 0.01 \\ p(MT = \text{Eng}|Cnt = S) &= 0.7 & p(MT = \text{Scot}|Cnt = S) &= 0.3 & p(MT = \text{Wel}|Cnt = S) &= 0.0 \\ p(MT = \text{Eng}|Cnt = W) &= 0.6 & p(MT = \text{Scot}|Cnt = W) &= 0.0 & p(MT = \text{Wel}|Cnt = W) &= 0.4 \end{aligned} \quad (1.1.15)$$

From this we can form a joint distribution $p(Cnt, MT) = p(MT|Cnt)p(Cnt)$. This could be written as a 3×3 matrix with (say) rows indexed by country and columns indexed by Mother Tongue:

$$\begin{pmatrix} 0.95 \times 0.88 & 0.7 \times 0.08 & 0.6 \times 0.04 \\ 0.04 \times 0.88 & 0.3 \times 0.08 & 0.0 \times 0.04 \\ 0.01 \times 0.88 & 0.0 \times 0.08 & 0.4 \times 0.04 \end{pmatrix} = \begin{pmatrix} 0.836 & 0.056 & 0.024 \\ 0.0352 & 0.024 & 0 \\ 0.0088 & 0 & 0.016 \end{pmatrix} \quad (1.1.16)$$

The joint distribution contains all the information about the model of this environment. By summing a column of this table, we have the marginal $p(Cnt)$. Summing the row gives the marginal $p(MT)$. Similarly, one could easily infer $p(Cnt|MT) \propto p(Cnt|MT)p(MT)$ from this joint distribution.

For joint distributions over a larger number of variables, $x_i, i = 1, \dots, D$, with each variable x_i taking K_i states, the table describing the joint distribution is an array with $\prod_{i=1}^D K_i$ entries. Explicitly storing tables therefore requires space exponential in the number of variables, which rapidly becomes impractical for a large number of variables.

A probability distribution assigns a value to each of the joint states of the variables. For this reason, $p(T, J, R, S)$ is considered equivalent to $p(J, S, R, T)$ (or any such reordering of the variables), since in each case the joint setting of the variables is simply a different index to the same probability. This situation is more clear in the set theoretic notation $p(J \cap S \cap T \cap R)$. We abbreviate this set theoretic notation by using the commas – however, one should be careful not to confuse the use of this indexing type notation with functions $f(x, y)$ which are in general dependent on the variable order. Whilst the variables to the left of the conditioning bar may be written in any order, and equally those to the right of the conditioning bar may be written in any order, moving variables across the bar is not generally equivalent, so that $p(x_1|x_2) \neq p(x_2|x_1)$.

1.1.2 Interpreting Conditional Probability

Together with the rules of probability, conditional probability enables one to reason in a rational, logical and consistent way. One could argue that much of science deals with problems of the form : tell me something about the parameters θ given that I have observed data $\mathcal{D}$ and have some knowledge of the underlying data generating mechanism. From a modelling perspective, this requires

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})} = \frac{p(\mathcal{D}|\theta)p(\theta)}{\int_{\theta} p(\mathcal{D}|\theta)p(\theta)} \quad (1.1.17)$$

This shows how from a forward or *generative model* $p(\mathcal{D}|\theta)$ of the dataset, and coupled with a *prior* belief $p(\theta)$ about which parameter values are appropriate, we can infer the *posterior* distribution $p(\theta|\mathcal{D})$ of parameters in light of the observed data.

This use of a generative model sits well with physical models of the world which typically postulate how to generate observed phenomena, assuming we know the correct parameters of the model. For example, one might postulate how to generate a time-series of displacements for a swinging pendulum but with unknown mass, length and damping constant. Using this generative model, and given only the displacements, we could infer the unknown physical properties of the pendulum, such as its mass, length and friction damping constant.

Subjective Probability

Probability is a contentious topic and we do not wish to get bogged down by the debate here, apart from pointing out that it is not necessarily the axioms of probability that are contentious rather what interpretation we should place on them. In some cases potential repetitions of an experiment can be envisaged so that the ‘long run’ (or frequentist) definition of probability in which probabilities are defined with respect to a potentially infinite repetition of ‘experiments’ makes sense. For example, in coin tossing, the probability of heads might be interpreted as ‘If I were to repeat the experiment of flipping a coin (at ‘random’), the limit of the number of heads that occurred over the number of tosses is defined as the probability of a head occurring.

Here’s another problem that is typical of the kind of scenario one might face in a machine learning situation. A film enthusiast joins a new online film service. Based on expressing a few films a user likes and dislikes, the online company tries to estimate the probability that the user will like each of the 10000 films in their database. If we were to define probability as a limiting case of infinite repetitions of the same experiment, this wouldn’t make much sense in this case since we can’t repeat the experiment. However, if we assume that the user behaves in a manner consistent with other users, we should be able to exploit the large amount of data from other users’ ratings to make a reasonable ‘guess’ as to what this consumer likes. This *degree of belief* or *Bayesian* subjective interpretation of probability sidesteps non-repeatability issues – it’s just a consistent framework for manipulating real values consistent with our intuition about probability[145].

1.2 Probabilistic Reasoning

The axioms of probability, combined with Bayes’ rule make for a complete reasoning system, one which includes traditional deductive logic as a special case[145].

Remark 1. The central paradigm of probabilistic reasoning is to identify all relevant variables $x_1, \dots, x_N$ in the environment, and make a probabilistic model $p(x_1, \dots, x_N)$ of their interaction. Reasoning (inference) is then performed by introducing *evidence*² that sets variables in known states, and subsequently computing probabilities of interest, conditioned on this evidence.

Example 1 (Hamburgers). Consider the following fictitious scientific information: Doctors find that people with Kreuzfeld-Jacob disease (KJ) almost invariably ate hamburgers, thus $p(\text{Hamburger Eater} | KJ) = 0.9$. The probability of an individual having *KJ* is currently rather low, about one in 100,000.

1. Assuming eating lots of hamburgers is rather widespread, say $p(\text{Hamburger Eater}) = 0.5$, what is the probability that a hamburger eater will have Kreuzfeld-Jacob disease?

This may be computed as

$$p(KJ | \text{Hamburger Eater}) = \frac{p(\text{Hamburger Eater}, KJ)}{p(\text{Hamburger Eater})} = \frac{p(\text{Hamburger Eater} | KJ)p(KJ)}{p(\text{Hamburger Eater})} \quad (1.2.1)$$

$$= \frac{\frac{9}{10} \times \frac{1}{100000}}{\frac{1}{2}} = 1.8 \times 10^{-5} \quad (1.2.2)$$

2. If the fraction of people eating hamburgers was rather small, $p(\text{Hamburger Eater}) = 0.001$, what is the probability that a regular hamburger eater will have Kreuzfeld-Jacob disease? Repeating the above calculation, this is given by

$$\frac{\frac{9}{10} \times \frac{1}{100000}}{\frac{1}{1000}} \approx 1/100 \quad (1.2.3)$$

Intuitively, this is much higher than in scenario (1) since here we can be more sure that eating hamburgers is related to the illness. In this case only a small number of people in the population eat hamburgers, and most of them get ill.

Example 2 (Inspector Clouseau). Inspector Clouseau arrives at the scene of a crime. The victim lies dead in the room and the inspector quickly finds the murder weapon, a Knife (K). The Butler (B) and Maid (M) are his main suspects. The inspector has a prior belief of 0.8 that the Butler is the murderer, and a prior belief of 0.2 that the Maid is the murderer. These probabilities are independent in the sense that $p(B, M) = p(B)p(M)$. (It is possible that both the Butler and the Maid murdered the victim or neither). The inspector's *prior* criminal knowledge can be formulated mathematically as follows:

$$\text{dom}(B) = \text{dom}(M) = \{\text{murderer, not murderer}\}, \text{dom}(K) = \{\text{knife used, knife not used}\} \quad (1.2.4)$$

$$p(B = \text{murderer}) = 0.8, \quad p(M = \text{murderer}) = 0.2 \quad (1.2.5)$$

$$\begin{aligned} p(\text{knife used} | B = \text{not murderer}, M = \text{not murderer}) &= 0.3 \\ p(\text{knife used} | B = \text{not murderer}, M = \text{murderer}) &= 0.2 \\ p(\text{knife used} | B = \text{murderer}, M = \text{not murderer}) &= 0.6 \\ p(\text{knife used} | B = \text{murderer}, M = \text{murderer}) &= 0.1 \end{aligned} \quad (1.2.6)$$

What is the probability that the Butler is the murderer? (Remember that it might be that neither is the murderer). Using b for the two states of B and m for the two states of M ,

$$p(B|K) = \sum_m p(B, m|K) = \sum_m \frac{p(B, m, K)}{p(K)} = \frac{p(B) \sum_m p(K|B, m)p(m)}{\sum_b p(b) \sum_m p(K|b, m)p(m)} \quad (1.2.7)$$

Plugging in the values we have

$$p(B = \text{murderer} | \text{knife used}) = \frac{\frac{8}{10} \left(\frac{2}{10} \times \frac{1}{10} + \frac{8}{10} \times \frac{6}{10} \right)}{\frac{8}{10} \left(\frac{2}{10} \times \frac{1}{10} + \frac{8}{10} \times \frac{6}{10} \right) + \frac{2}{10} \left(\frac{2}{10} \times \frac{2}{10} + \frac{8}{10} \times \frac{3}{10} \right)} = \frac{200}{228} \approx 0.877 \quad (1.2.8)$$

The role of $p(\text{knife used})$ in the Inspector Clouseau example can cause some confusion. In the above,

$$p(\text{knife used}) = \sum_b p(b) \sum_m p(\text{knife used} | b, m)p(m) \quad (1.2.9)$$

is computed to be 0.456. But surely, $p(\text{knife used}) = 1$, since this is given in the question! Note that the quantity $p(\text{knife used})$ relates to the *prior* probability the model assigns to the knife being used (in the absence of any other information). If we know that the knife is used, then the *posterior*

$$p(\text{knife used} | \text{knife used}) = \frac{p(\text{knife used}, \text{knife used})}{p(\text{knife used})} = \frac{p(\text{knife used})}{p(\text{knife used})} = 1 \quad (1.2.10)$$

which, naturally, must be the case.

Another potential confusion is the choice

$$p(B = \text{murderer}) = 0.8, \quad p(M = \text{murderer}) = 0.2 \quad (1.2.11)$$

which means that $p(B = \text{not murderer}) = 0.2$, $p(M = \text{not murderer}) = 0.8$. These events are not exclusive and it's just 'coincidence' that the numerical values are chosen this way. For example, we could have also chosen

$$p(B = \text{murderer}) = 0.6, \quad p(M = \text{murderer}) = 0.9 \quad (1.2.12)$$

which means that $p(B = \text{not murderer}) = 0.4$, $p(M = \text{not murderer}) = 0.1$

1.3 Prior, Likelihood and Posterior

The prior, likelihood and posterior are all probabilities. They are assigned these names due to their role in Bayes' rule, described below.

Definition 7. Prior Likelihood and Posterior

For data $\mathcal{D}$ and variable θ , Bayes' rule tells us how to update our prior beliefs about the variable θ in light of the data to a posterior belief:

$$\underbrace{p(\theta|\mathcal{D})}_{\text{posterior}} = \frac{\underbrace{p(\mathcal{D}|\theta)}_{\text{likelihood}} \underbrace{p(\theta)}_{\text{prior}}}{\underbrace{p(\mathcal{D})}_{\text{evidence}}} \quad (1.3.1)$$

The *evidence* is also called the *marginal likelihood*.

The term likelihood is used for the probability that a model generates observed data. More fully, if we condition on the model M , we have

$$p(\theta|\mathcal{D}, M) = \frac{p(\mathcal{D}|\theta, M)p(\theta|M)}{p(\mathcal{D}|M)}$$

where we see the role of the likelihood $p(\mathcal{D}|\theta, M)$ and marginal likelihood $p(\mathcal{D}|M)$. The marginal likelihood is also called the *model likelihood*.

The *most probable a posteriori* (*MAP*) setting is that which maximises the posterior, $\theta_* = \underset{\theta}{\operatorname{argmax}} p(\theta|\mathcal{D}, M)$.

Bayes' rule tells us how to update our prior knowledge with the data generating mechanism. The prior distribution $p(\theta)$ describes the information we have about the variable before seeing any data. After data $\mathcal{D}$ arrives, we update the prior distribution to the posterior $p(\theta|\mathcal{D}) \propto p(\mathcal{D}|\theta)p(\theta)$.

1.3.1 Two dice : what were the individual scores?

Two fair dice are rolled. Someone tells you that the sum of the two scores is 9. What is the probability distribution of the two dice scores³?

The score of die a is denoted s_a with $\operatorname{dom}(s_a) = \{1, 2, 3, 4, 5, 6\}$ and similarly for s_b . The three variables involved are then s_a , s_b and the total score, $t = s_a + s_b$. A model of these three variables naturally takes the form

$$p(t, s_a, s_b) = \underbrace{p(t|s_a, s_b)}_{\text{likelihood}} \underbrace{p(s_a, s_b)}_{\text{prior}} \quad (1.3.2)$$

The prior $p(s_a, s_b)$ is the joint probability of score s_a and score s_b without knowing anything else. Assuming no dependency in the rolling mechanism,

$$p(s_a, s_b) = p(s_a)p(s_b) \quad (1.3.3)$$

Since the dice are fair both $p(s_a)$ and $p(s_b)$ are uniform distributions, $p(s_a = s) = 1/6$.

$p(s_a)p(s_b)$:

	$s_a = 1$	$s_a = 2$	$s_a = 3$	$s_a = 4$	$s_a = 5$	$s_a = 6$
$s_b = 1$	1/36	1/36	1/36	1/36	1/36	1/36
$s_b = 2$	1/36	1/36	1/36	1/36	1/36	1/36
$s_b = 3$	1/36	1/36	1/36	1/36	1/36	1/36
$s_b = 4$	1/36	1/36	1/36	1/36	1/36	1/36
$s_b = 5$	1/36	1/36	1/36	1/36	1/36	1/36
$s_b = 6$	1/36	1/36	1/36	1/36	1/36	1/36

³This example is due to Taylan Cemgil.

Here the likelihood term is

$$p(t|s_a, s_b) = \mathbb{I}[t = s_a + s_b] \quad (1.3.4)$$

which states that the total score is given by $s_a + s_b$. Here $\mathbb{I}[x = y]$ is the *indicator function* defined as $\mathbb{I}[x = y] = 1$ if $x = y$ and 0 otherwise.

$$p(t = 9|s_a, s_b):$$

	$s_a = 1$	$s_a = 2$	$s_a = 3$	$s_a = 4$	$s_a = 5$	$s_a = 6$
$s_b = 1$	0	0	0	0	0	0
$s_b = 2$	0	0	0	0	0	0
$s_b = 3$	0	0	0	0	0	1
$s_b = 4$	0	0	0	0	1	0
$s_b = 5$	0	0	0	1	0	0
$s_b = 6$	0	0	1	0	0	0

Hence, our complete model is

$$p(t, s_a, s_b) = p(t|s_a, s_b)p(s_a)p(s_b) \quad (1.3.5)$$

where the terms on the right are explicitly defined.

Our interest is then obtainable using Bayes' rule,

$$p(s_a, s_b|t = 9) = \frac{p(t = 9|s_a, s_b)p(s_a)p(s_b)}{p(t = 9)} \quad (1.3.6)$$

where

$$p(t = 9) = \sum_{s_a, s_b} p(t = 9|s_a, s_b)p(s_a)p(s_b) \quad (1.3.7)$$

$$p(t = 9|s_a, s_b)p(s_a)p(s_b):$$

	$s_a = 1$	$s_a = 2$	$s_a = 3$	$s_a = 4$	$s_a = 5$	$s_a = 6$
$s_b = 1$	0	0	0	0	0	0
$s_b = 2$	0	0	0	0	0	0
$s_b = 3$	0	0	0	0	0	1/36
$s_b = 4$	0	0	0	0	1/36	0
$s_b = 5$	0	0	0	1/36	0	0
$s_b = 6$	0	0	1/36	0	0	0

$$p(s_a, s_b|t = 9):$$

	$s_a = 1$	$s_a = 2$	$s_a = 3$	$s_a = 4$	$s_a = 5$	$s_a = 6$
$s_b = 1$	0	0	0	0	0	0
$s_b = 2$	0	0	0	0	0	0
$s_b = 3$	0	0	0	0	0	1/4
$s_b = 4$	0	0	0	0	1/4	0
$s_b = 5$	0	0	0	1/4	0	0
$s_b = 6$	0	0	1/4	0	0	0

The term $p(t = 9) = \sum_{s_a, s_b} p(t = 9|s_a, s_b)p(s_a)p(s_b) = 4 \times 1/36 = 1/9$. Hence the posterior is given by equal mass in only 4 non-zero elements, as shown.

1.4 Further worked examples

Example 3 (Who's in the bathroom?). Consider a household of three people, Alice, Bob and Cecil. Cecil wants to go to the bathroom but finds it occupied. He then goes to Alice's room and sees she is there. Since Cecil knows that only either Alice or Bob can be in the bathroom, from this he infers that Bob must be in the bathroom.

To arrive at the same conclusion in a mathematical framework, let's define the following events

$$A = \text{Alice is in her bedroom}, \quad B = \text{Bob is in his bedroom}, \quad O = \text{Bathroom occupied} \quad (1.4.1)$$

We can encode the information that if either Alice or Bob are not in their bedrooms, then they must be in the bathroom (they might both be in the bathroom) as

$$p(O = \text{tr}|A = \text{fa}, B) = 1, \quad p(O = \text{tr}|A, B = \text{fa}) = 1 \quad (1.4.2)$$

The first term expresses that the bathroom is occupied if Alice is not in her bedroom, wherever Bob is. Similarly, the second term expresses bathroom occupancy as long as Bob is not in his bedroom. Then

$$p(B = \text{fa}|O = \text{tr}, A = \text{tr}) = \frac{p(B = \text{fa}, O = \text{tr}, A = \text{tr})}{p(O = \text{tr}, A = \text{tr})} = \frac{p(O = \text{tr}|A = \text{tr}, B = \text{fa})p(A = \text{tr}, B = \text{fa})}{p(O = \text{tr}, A = \text{tr})} \quad (1.4.3)$$

where

$$p(O = \text{tr}, A = \text{tr}) = p(O = \text{tr}|A = \text{tr}, B = \text{fa})p(A = \text{tr}, B = \text{fa}) + p(O = \text{tr}|A = \text{tr}, B = \text{tr})p(A = \text{tr}, B = \text{tr}) \quad (1.4.4)$$

Using the fact $p(O = \text{tr}|A = \text{tr}, B = \text{fa}) = 1$ and $p(O = \text{tr}|A = \text{tr}, B = \text{tr}) = 0$, which encodes that if Alice is in her room and Bob is not, the bathroom must be occupied, and similarly, if both Alice and Bob are in their rooms, the bathroom cannot be occupied,

$$p(B = \text{fa}|O = \text{tr}, A = \text{tr}) = \frac{p(A = \text{tr}, B = \text{fa})}{p(A = \text{tr}, B = \text{fa})} = 1 \quad (1.4.5)$$

This example is interesting since we are not required to make a full probabilistic model in this case thanks to the limiting nature of the probabilities (we don't need to specify $p(A, B)$). The situation is common in limiting situations of probabilities being either 0 or 1, corresponding to traditional logic systems.

Example 4 (Aristotle : Resolution). We can represent the statement 'All apples are fruit' by $p(F = \text{tr}|A = \text{tr}) = 1$. Similarly, 'All fruits grow on trees' may be represented by $p(T = \text{tr}|F = \text{tr}) = 1$. Additionally we assume that whether or not something grows on a tree depends only on whether or not it is a fruit, $p(T|A, F) = P(T|F)$. From these, we can compute

$$p(T = \text{tr}|A = \text{tr}) = \sum_F p(T = \text{tr}|F, A = \text{tr})p(F|A = \text{tr}) = \sum_F p(T = \text{tr}|F)p(F|A = \text{tr}) \quad (1.4.6)$$

$$= p(T = \text{tr}|F = \text{fa}) \underbrace{p(F = \text{fa}|A = \text{tr})}_{=0} + \underbrace{p(T = \text{tr}|F = \text{tr})}_{=1} \underbrace{p(F = \text{tr}|A = \text{tr})}_{=1} = 1 \quad (1.4.7)$$

In other words we have deduced that 'All apples grow on trees' is a true statement, based on the information presented. (This kind of reasoning is called resolution and is a form of transitivity : from the statements $A \Rightarrow F$ and $F \Rightarrow T$ we can infer $A \Rightarrow T$).

Example 5 (Aristotle : Inverse Modus Ponens). According to Logic, from the statement : 'If A is true then B is true', one may deduce that 'if B is false then A is false'. Let's see how this fits in with a probabilistic reasoning system. We can express the statement : 'If A is true then B is true' as $p(B = \text{tr}|A = \text{tr}) = 1$. Then we may infer

$$\begin{aligned} p(A = \text{fa}|B = \text{fa}) &= 1 - p(A = \text{tr}|B = \text{fa}) \\ &= 1 - \frac{p(B = \text{fa}|A = \text{tr})p(A = \text{tr})}{p(B = \text{fa}|A = \text{tr})p(A = \text{tr}) + p(B = \text{fa}|A = \text{fa})p(A = \text{fa})} = 1 \end{aligned} \quad (1.4.8)$$

This follows since $p(B = \text{fa}|A = \text{tr}) = 1 - p(B = \text{tr}|A = \text{tr}) = 1 - 1 = 0$, annihilating the second term.

Both the above examples are intuitive expressions of deductive logic. The standard rules of Aristotelian logic are therefore seen to be limiting cases of probabilistic reasoning.

Example 6 (Soft XOR Gate).

A standard XOR logic gate is given by the table on the right. If we observe that the output of the XOR gate is 0, what can we say about A and B ? In this case, either A and B were both 0, or A and B were both 1. This means we don't know which state A was in – it could equally likely have been 1 or 0.

A	B	$A \text{ xor } B$
0	0	0
0	1	1
1	0	1
1	1	0

Consider a ‘soft’ version of the XOR gate given on the right, with additionally $p(A = 1) = 0.65$, $p(B = 1) = 0.77$. What is $p(A = 1|C = 0)$?

A	B	$p(C = 1 A, B)$
0	0	0.1
0	1	0.99
1	0	0.8
1	1	0.25

$$\begin{aligned}
 p(A = 1, C = 0) &= \sum_B p(A = 1, B, C = 0) = \sum_B p(C = 0|A = 1, B)p(A = 1)p(B) \\
 &= p(A = 1) (p(C = 0|A = 1, B = 0)p(B = 0) + p(C = 0|A = 1, B = 1)p(B = 1)) \\
 &= 0.65 \times (0.2 \times 0.23 + 0.75 \times 0.77) = 0.4053
 \end{aligned} \tag{1.4.9}$$

$$\begin{aligned}
 p(A = 0, C = 0) &= \sum_B p(A = 0, B, C = 0) = \sum_B p(C = 0|A = 0, B)p(A = 0)p(B) \\
 &= p(A = 0) (p(C = 0|A = 0, B = 0)p(B = 0) + p(C = 0|A = 0, B = 1)p(B = 1)) \\
 &= 0.35 \times (0.9 \times 0.23 + 0.01 \times 0.77) = 0.0751
 \end{aligned}$$

Then

$$p(A = 1|C = 0) = \frac{p(A = 1, C = 0)}{p(A = 1, C = 0) + p(A = 0, C = 0)} = \frac{0.4053}{0.4053 + 0.0751} = 0.8437 \tag{1.4.10}$$

Example 7 (Larry). Larry is typically late for school. If Larry is late, we denote this with $L = \text{late}$, otherwise, $L = \text{not late}$. When his mother asks whether or not he was late for school he never admits to being late. The response Larry gives R_L is represented as follows

$$p(R_L = \text{not late}|L = \text{not late}) = 1, \quad p(R_L = \text{late}|L = \text{late}) = 0 \tag{1.4.11}$$

The remaining two values are determined by normalisation and are

$$p(R_L = \text{late}|L = \text{not late}) = 0, \quad p(R_L = \text{not late}|L = \text{late}) = 1 \tag{1.4.12}$$

Given that $R_L = \text{not late}$, what is the probability that Larry was late, *i.e.* $p(L = \text{late}|R_L = \text{not late})$?

Using Bayes’ we have

$$\begin{aligned}
 p(L = \text{late}|R_L = \text{not late}) &= \frac{p(L = \text{late}, R_L = \text{not late})}{p(R_L = \text{not late})} \\
 &= \frac{p(L = \text{late}, R_L = \text{not late})}{p(L = \text{late}, R_L = \text{not late}) + p(L = \text{not late}, R_L = \text{not late})}
 \end{aligned} \tag{1.4.13}$$

In the above

$$p(L = \text{late}, R_L = \text{not late}) = \underbrace{p(R_L = \text{not late}|L = \text{late})}_{=1} p(L = \text{late}) \tag{1.4.14}$$

and

$$p(L = \text{not late}, R_L = \text{not late}) = \underbrace{p(R_L = \text{not late}|L = \text{not late})}_{=1} p(L = \text{not late}) \tag{1.4.15}$$

Hence

$$p(L = \text{late}|R_L = \text{not late}) = \frac{p(L = \text{late})}{p(L = \text{late}) + p(L = \text{not late})} = p(L = \text{late}) \tag{1.4.16}$$

Where we used normalisation in the last step, $p(L = \text{late}) + p(L = \text{not late}) = 1$. This result is intuitive – Larry’s mother knows that he never admits to being late, so her belief about whether or not he really was late is unchanged, regardless of what Larry actually says.

Example 8 (Larry and Sue). Continuing the example above, Larry’s sister Sue always tells the truth to her mother as to whether or not Larry was late for School.

$$p(R_S = \text{not late} | L = \text{not late}) = 1, \quad p(R_S = \text{late} | L = \text{late}) = 1 \quad (1.4.17)$$

The remaining two values are determined by normalisation and are

$$p(R_S = \text{late} | L = \text{not late}) = 0, \quad p(R_S = \text{not late} | L = \text{late}) = 0 \quad (1.4.18)$$

We also assume $p(R_S, R_L | L) = p(R_S | L)p(R_L | L)$. We can then write

$$p(R_L, R_S, L) = p(R_L | L)p(R_S | L)p(L) \quad (1.4.19)$$

Given that $R_S = \text{late}$, what is the probability that Larry was late?

Using Bayes’ rule, we have

$$p(L = \text{late} | R_L = \text{not late}, R_S = \text{late}) = \frac{1}{Z} p(R_S = \text{late} | L = \text{late}) p(R_L = \text{not late} | L = \text{late}) p(L = \text{late})$$

where the normalisation Z is given by

$$\begin{aligned} & p(R_S = \text{late} | L = \text{late}) p(R_L = \text{not late} | L = \text{late}) p(L = \text{late}) \\ & + p(R_S = \text{late} | L = \text{not late}) p(R_L = \text{not late} | L = \text{not late}) p(L = \text{not late}) \end{aligned}$$

Hence

$$p(L = \text{late} | R_L = \text{not late}, R_S = \text{late}) = \frac{1 \times 1 \times p(L = \text{late})}{1 \times 1 \times p(L = \text{late}) + 0 \times 1 \times p(L = \text{not late})} = 1 \quad (1.4.20)$$

This result is also intuitive – Since Larry’s mother knows that Sue always tells the truth, no matter what Larry says, she knows he was late.

Example 9 (Luke). Luke has been told he’s lucky and has won a prize in the lottery. There are 5 prizes available of value £10, £100, £1000, £10000, £1000000. The prior probabilities of winning these 5 prizes are p_1, p_2, p_3, p_4, p_5 , with p_0 being the prior probability of winning no prize. Luke asks eagerly ‘Did I win £1000000?!’. ‘I’m afraid not sir’, is the response of the lottery phone operator. ‘Did I win £10000?!’ asks Luke. ‘Again, I’m afraid not sir’. What is the probability that Luke has won £1000?

Note first that $p_0 + p_1 + p_2 + p_3 + p_4 + p_5 = 1$. We denote $W = 1$ for the first prize of £10, and $W = 2, \dots, 5$ for the remaining prizes and $W = 0$ for no prize. We need to compute

$$\begin{aligned} p(W = 3 | W \neq 5, W \neq 4, W \neq 0) &= \frac{p(W = 3, W \neq 5, W \neq 4, W \neq 0)}{p(W \neq 5, W \neq 4, W \neq 0)} \\ &= \frac{p(W = 3)}{p(W = 1 \text{ or } W = 2 \text{ or } W = 3)} = \frac{p_3}{p_1 + p_2 + p_3} \end{aligned} \quad (1.4.21)$$

where the term in the denominator is computed using the fact that the events W are mutually exclusive (one can only win one prize). This result makes intuitive sense : once we have removed the impossible states of W , the probability that Luke wins the prize is proportional to the prior probability of that prize, with the normalisation being simply the total set of possible probability remaining.

1.5 Code

The BRMLTOOLBOX code accompanying this book is intended to give the reader some insight into representing discrete probability tables and performing simple inference. The MATLAB code is written with only minimal error trapping to keep the code as short and hopefully reasonably readable⁴.

1.5.1 Basic Probability code

At the simplest level, we only need two basic routines. One for multiplying probability tables together (called potentials in the code), and one for summing a probability table. Potentials are represented using a structure. For example, in the code corresponding to the Inspector Clouseau example `demoClouseau.m`, we define a probability table as

```
>> pot(1)
ans =
    variables: [1 3 2]
    table: [2x2x2 double]
```

This says that the potential depends on the variables 1,3,2 and the entries are stored in the array given by the `table` field. The size of the array informs how many states each variable takes in the order given by `variables`. The order in which the variables are defined in a potential is irrelevant provided that one indexes the array consistently. A routine that can help with setting table entries is `setstate.m`. For example,

```
>> pot(1) = setstate(pot(1),[2 1 3],[2 1 1],0.3)
```

means that for potential 1, the table entry for variable 2 being in state 2, variable 1 being in state 1 and variable 3 being in state 1 should be set to value 0.3.

The philosophy of the code is to keep the information required to perform computations to a minimum. Additional information about the labels of variables and their domains can be useful to check results, but is not actually required to carry out computations. One may also specify the name and domain of each variable, for example

```
>>variable(3)
ans =
    domain: {'murderer' 'not murderer'}
    name: 'butler'
```

The variable name and domain information in the Clouseau example is stored in the structure `variable`, which can be helpful to display the potential table:

```
>> disptable(pot(1),variable);
knife = used      maid = murderer      butler = murderer      0.100000
knife = not used   maid = murderer      butler = murderer      0.900000
knife = used      maid = not murderer    butler = murderer      0.600000
knife = not used   maid = not murderer    butler = murderer      0.400000
knife = used      maid = murderer      butler = not murderer  0.200000
knife = not used   maid = murderer      butler = not murderer  0.800000
knife = used      maid = not murderer    butler = not murderer  0.300000
knife = not used   maid = not murderer    butler = not murderer  0.700000
```

⁴At the time of writing, some versions of MATLAB suffer from serious memory indexing bugs, some of which may appear in the array structures used in the code provided. To deal with this, turn off the JIT accelerator by typing `feature accel off`.

Multiplying Potentials

In order to multiply potentials (as for arrays) the tables of each potential must be dimensionally consistent – that is the number of states of variable i in potential 1 must match the number of states of variable i in any other potential. This can be checked using `potvariables.m`. This consistency is also required for other basic operations such as summing potentials.

`multpots.m`: Multiplying two or more potentials

`divpots.m`: Dividing a potential by another

Summing a Potential

`sumpot.m`: Sum (marginalise) a potential over a set of variables

`sumpots.m`: Sum a set of potentials together

Making a conditional Potential

`condpot.m`: Make a potential conditioned on variables

Setting a Potential

`setpot.m`: Set variables in a potential to given states

`setevpot.m`: Set variables in a potential to given states and return also an identity potential on the given states

The philosophy of `BRMLTOOLBOX` is that all information about variables is local and is read off from a potential. Using `setevpot.m` enables one to set variables in a state whilst maintaining information about the number of states of a variable.

Maximising a Potential

`maxpot.m`: Maximise a potential over a set of variables

See also `maxNarray.m` and `maxNpot.m` which return the N -highest values and associated states.

Other potential utilities

`setstate.m`: Set the a potential state to a given value

`table.m`: Return a table from a potential

`whichpot.m`: Return potentials which contain a set of variables

`potvariables.m`: Variables and their number of states in a set of potentials

`orderpotfields.m`: Order the fields of a potential structure

`uniquepots.m`: Merge redundant potentials and return only unique ones

`numstates.m`: Number of states of a variable in a domain

`squeezepots.m`: Remove redundant potentials by merging

`normpot.m`: Normalise a potential to form a distribution

1.5.2 General utilities

`condp.m`: Return a table $p(x|y)$ from $p(x, y)$

`condexp.m`: Form a conditional distribution from a log value

`logsumexp.m`: Compute the log of a sum of exponentials in a numerically precise way

`normp.m`: Return a normalised table from an unnormalised table

`assign.m`: Assign values to multiple variables

`maxarray.m`: Maximize a multi-dimensional array over a subset

1.5.3 An example

The following code highlights the use of the above routines in solving the Inspector Clouseau, example(2).

`demoClouseau.m`: Solving the Inspector Clouseau example

1.6 Notes

The interpretation of probability is contentious and we refer the reader to [145, 183, 179] for detailed discussions. A useful website that relates to understanding probability and Bayesian reasoning is understandinguncertainty.org

1.7 Exercises

Exercise 1. *Prove*

$$p(x, y|z) = p(x|z)p(y|x, z) \quad (1.7.1)$$

and also

$$p(x|y, z) = \frac{p(y|x, z)p(x|z)}{p(y|z)} \quad (1.7.2)$$

Exercise 2. *Prove the Bonferroni inequality*

$$p(a, b) \geq p(a) + p(b) - 1 \quad (1.7.3)$$

Exercise 3 (Adapted from [167]). *There are two boxes. Box 1 contains three red and five white balls and box 2 contains two red and five white balls. A box is chosen at random $p(\text{box} = 1) = p(\text{box} = 2) = 0.5$ and a ball chosen at random from this box turns out to be red. What is the posterior probability that the red ball came from box 1?*

Exercise 4 (Adapted from [167]). *Two balls are placed in a box as follows: A fair coin is tossed and a white ball is placed in the box if a head occurs, otherwise a red ball is placed in the box. The coin is tossed again and a red ball is placed in the box if a tail occurs, otherwise a white ball is placed in the box. Balls are drawn from the box three times in succession (always with replacing the drawn ball back in the box). It is found that on all three occasions a red ball is drawn. What is the probability that both balls in the box are red?*

Exercise 5 (From David Spiegelhalter understandinguncertainty.org). *A secret government agency has developed a scanner which determines whether a person is a terrorist. The scanner is fairly reliable; 95% of all scanned terrorists are identified as terrorists, and 95% of all upstanding citizens are identified as such. An informant tells the agency that exactly one passenger of 100 aboard an aeroplane in which you are seated is a terrorist. The agency decide to scan each passenger and the shifty looking man sitting next to you is the first to test positive. What are the chances that this man is a terrorist?*

Exercise 6 (The Monty Hall problem). *On a gameshow there are three doors. Behind one door is a prize. The gameshow host asks you to pick a door. He then opens a different door to the one you chose and shows that there is no prize behind it. Is it better to stick with your original guess as to where the prize is, or better to change your mind?*

Exercise 7. *Consider three variable distributions which admit the factorisation*

$$p(a, b, c) = p(a|b)p(b|c)p(c) \quad (1.7.4)$$

where all variables are binary. How many parameters are needed to specify distributions of this form?

Exercise 8. Repeat the Inspector Clouseau scenario, example(2), but with the restriction that either the Maid or the Butler is the murderer, but not both. Explicitly, the probability of the Maid being the Murderer and not the Butler is 0.04, the probability of the Butler being the Murderer and not the Maid is 0.64. Modify `demoClouseau.m` to implement this.

Exercise 9. Prove

$$p(a, (b \text{ or } c)) = p(a, b) + p(a, c) - p(a, b, c) \quad (1.7.5)$$

Exercise 10. Prove

$$p(x|z) = \sum_y p(x|y, z)p(y|z) = \sum_{y,w} p(x|w, y, z)p(w|y, z)p(y|z) \quad (1.7.6)$$

Exercise 11. As a young man Mr Gott visits Berlin in 1969. He's surprised that he cannot cross into East Berlin since there is a wall separating the two halves of the city. He's told that the wall was erected 8 years previously. He reasons that : The wall will have a finite lifespan; his ignorance means that he arrives uniformly at random at some time in the lifespan of the wall. Since only 5% of the time one would arrive in the first or last 2.5% of the lifespan of the wall he asserts that with 95% confidence the wall will survive between $8/0.975 \approx 8.2$ and $8/0.025 = 320$ years. In 1989 the now Professor Gott is pleased to find that his prediction was correct and promotes his prediction method in elite journals. This 'delta-t' method is widely adopted and used to form predictions in a range of scenarios about which researchers are 'totally ignorant'. Would you 'buy' a prediction from Prof. Gott? Explain carefully your reasoning.

Exercise 12. Implement the soft XOR gate, example(6) using BRMLTOOLBOX. You may find `condpot.m` of use.

Exercise 13. Implement the hamburgers, example(1) (both scenarios) using BRMLTOOLBOX. To do so you will need to define the joint distribution $p(\text{hamburgers}, KJ)$ in which $\text{dom}(\text{hamburgers}) = \text{dom}(KJ) = \{\text{tr}, \text{fa}\}$.

Exercise 14. Implement the two-dice example, section(1.3.1) using BRMLTOOLBOX.

Exercise 15. A redistribution lottery involves picking the correct four numbers from 1 to 9 (without replacement, so 3,4,4,1 for example is not possible). The order of the picked numbers is irrelevant. Every week a million people play this game, each paying £1 to enter, with the numbers 3,5,7,9 being the most popular (1 in every 100 people chooses these numbers). Given that the million pounds prize money is split equally between winners, and that any four (different) numbers come up at random, what is the expected amount of money each of the players choosing 3,5,7,9 will win each week? The least popular set of numbers is 1,2,3,4 with only 1 in 10,000 people choosing this. How much do they profit each week, on average? Do you think there is any 'skill' involved in playing this lottery?

Exercise 16. In a test of 'psychometry' the car keys and wrist watches of 5 people are given to a medium. The medium then attempts to match the wrist watch with the car key of each person. What is the expected number of correct matches that the medium will make (by chance)? What is the probability that the medium will obtain at least 1 correct match?